

TU Berlin Chair Agro Mechatronics Prof. Dr.-Ing. C. Weltzien				Requirements List for Modul Mobile Working Robotic Systems		Order No.: SoSe-2025-3-1 Team Members: I. Zytoon, N. Aschenbroich, J. Korn			
				Requirements		Specification – Data			
No.	Date:	Origin	D/R (F/W)			Minimum Fulfilment	STANDARD Fulfilment	Ideal Fulfilment	Unit [mm]
G01	28.04.25	C	D	General Requirements Robust and safe design Drivable omnidirectional robot platform Drive-by-Wire control Interface definition to/from navigation control Fitting, robust & cost-efficient selection of electrical components Design for agricultural application, in remote locations Use existing platform and motors (NEMA23 stepper, 57BLF03 BLDC) Compact, modular design for maintainability and debug access Completion of project until 16.07.25 , submission of documentation until 13.08.25					
G02	28.04.25	C	D						
G03	28.04.25	C	D						
G04	28.04.25	C	D						
G05	28.04.25	C	D						
G06	28.04.25	C	D						
G07	9.5.2025	IZ	D						
G08	9.5.2025	IZ	D						
G09	28.04.25	C	D						
				Specific Requirements (In Progress)					
S01	28.04.25	C	R	Remote control with MIT Inventor App via Bluetooth/WIFI (Raspberry Pi needed) Multiple drive modes with flexible remote mode switching Fail-safe for loss of connection (e.g. ROS ping messages) Integration of hardware-level safety functionality (Emergency Stop) ROS2 interface for future integration of advanced autonomy (e.g. path planning, mapping)		Holo- nomic, spin-in- place		Holonomic, spin-in-place, Ackermann	
S02	28.04.25	JK	D						
S03	28.04.25	JK	R						
S04	28.04.25	C	R						
S05	28.04.25	JK	R						
				Component Enclosure Requirements					
C01	09.05.25	JK	D	Robust enclosure for Raspberry Pi, STM32 board, converters, and interfaces 3D printed from ABS, modular layout for additional interface boards and sensors Screw cap, easy access for debugging and maintenance Cable entries with sealed grommets Heat management		LWH: 170x198x 166 [mm]			
C02	09.05.25	JK, IZ	R						
C03	09.05.25	JK	R						
C04	09.05.25	JK	R						
C05	09.05.25	JK	D						
				Process Requirements					
P01	28.04.25	C	D	Meet all deadlines and due dates Keep to the house rules Respectful and careful handling of facilities, infrastructure, tools in student working room Extensive and timely documentation of work and interface definition		Passive (e.g. vent holes)		Active (One or multiple fans)	
P02	28.04.25	C	D						
P03	28.04.25	C	D						
P04	28.04.25	JK	D						
Type of Requirements: D = Demands (F-Forderungen), R = Request (W-Wunsch, "nice-to-have"); C = Customer, JK = J. Korn, IZ = Iyhab Zytoon									
Replaces version:2 From:28.04.25 First version: Responsible:							Version: 3 Date: 09.05.2025 Page 1 of 1		